a)

Let D be a diagonal matrix of w_i i.e.

$$D = \begin{pmatrix} \sqrt{w_1} & & & 0 \\ & \sqrt{w_2} & & \\ & & \ddots & \\ 0 & & & \sqrt{w_m} \end{pmatrix}$$

Then DA gives us the row scaled matrix A where each row is multiplied by respective $\sqrt{w_i}$. Similarly when D is multiplied by b we get a vector of entries $\sqrt{w_i} \cdot b_i$. Thus

$$\|D(Ax-b)\| = \sum_{i=1}^{m} (\underbrace{\sqrt{w_i}\, a_i^T x}_{\text{rows of } DA} - \underbrace{\sqrt{w_i}\, b_i}_{\text{rows of } Db})^2 =$$

$$= \sum_{i=1}^{m} w_i (a_i^T x - b_i)^2$$

b) since we want to minimize $\|DAx - Db\|^2$ we expand it and get:

$$\varphi(x) = (DAx - Db)^T (DAx - Db) =$$

$$= x^T A^T D^T D A x - x^T A^T D^T D b - b^T D^T D A x -$$

$$- b^T D^T D b$$

We take The derivative with resp to x

$$0 = \varphi'_x(x) = (A^T D^T D A + A^T D^T D A)x - A^T D^T D b - b^T D^T D A$$

scalar → sam scalar's transpose. i.e. equal.

So $A^T \underbrace{D^T D}_{W} A x = A^T \underbrace{D^T D}_{W} b \Rightarrow x = (A^T W A)^{-1} A^T W b$